

Analytical Study of a Retail Accessories Shop

A Mid-Term report for the BDM capstone Project

Submitted by

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1 Executive Summary

'Sudhir Kumar Belts & Goggles' is a well-established accessory selling business with a 40-year legacy, located on GT Road, Azadpur , New Delhi. Operating alongside 20-25 vendors, the business works on a B2C model, offering high-quality products to consumers. Despite its longstanding presence and strong relationships within the market, the business faces challenges in removing dead stock from the shop and pricing strategies hindering its financial stability and growth potential.

The challenges faced by 'Sudhir Kumar Belts & Goggles' are excess inventory of outdated belts and goggles, which ties up capital and limits storage space, preventing the introduction of new trends and reducing profitability. Additionally, the shop's sales and revenue remain stagnant, with minimal growth across months. This lack of profitability and responsiveness to market demands restricts opportunities for expansion, innovation, and improved overall performance, leaving the business unable to capitalize on seasonal peaks or changes in customer demand.

To address these challenges, I propose a comprehensive analytical approach. By analyzing past sales and engaging in discussions with the business owner, I aim to identify performances of products and understand the financial landscape better. This will help me provide the financial overview of the business. Analyzing the details and preferences of customers will allow me to create a structured system to increase profitability.

2 Proof of originality of the Data

- Business Name : Sudhir Kumar Belts & Goggles
- Address : Gt Road , Azadpur , North Delhi - 110033
- Owner's Name : Sudhir Kumar

Video of Interaction with Business Owner : [Video of Interaction](#)

Letter : [Letter](#)



Fig 1. Me along with the Business Owner at the Business place

The business owner primarily uses two methods for Recording Data:

1. Khata Book : Used at the owner's residence for recording and calculating detailed monthly income from the shop.
2. Daily Notepad : Used to record the number of items sold per day.

2015	
2053	- 6
6084	- 3
2080	- 2
K-15	- 0
Paris	- 0
Spoke	- 7
Rav	- 1
Simple	2053 - 30
	Paris - 100
	Rav - 10
Wood	- 3
Khata	- 6
Cards	- 4

2053	- 57	x 30	- 1710
2080	- 70	x 25	- 1750
6084	- 62	x 28	- 1736
K-15	- 47	x 18	- 846
Paris	- 70	x 20	- 1400
Spoke	- 38	x 22	- 836
Rav	- 53	x 25	- 1325
			3853
			+ 150
			4003
			- Cards maintenance
			1200
			2803
2053	- 90	x 30	- 2700
Rav	- 12	x 10	- 120
Spoke	- 46	x 15	- 690
Paris	- 35	x 20	- 700
Wood	- 7	x 55	- 385
Khata	- 31	x 20	- 620
Cards	- 3	x 50	- 150
			3655
			3655 + 7200 = 11555

Fig 2. Snapshot of various Data Recordings

3 Metadata

- Data Format : Excel/Sheets (XLSX)
- Range : July 1, 2024, to August 31, 2024
- Business Closed : The business was closed on Sundays.
- Units of Measurement for Features involving Money : Indian Rupee (₹)

As the shop uses outdated methods for data entry and management, some figures might be approximate or slightly inaccurate due to manual calculation errors.

Information about the Sales Data of Business - [Sales Data](#)

Information Collected : This excel file contains 3 sheets.

→ Sheet 1 -

- Types of products in each category(Goggles & Belts).
- Cost Price : Cost Price incurred while purchasing items from different suppliers.
- Selling Price : The selling price for the items.
- Information about the suppliers and the brands they supply to the shop.

→ Sheet 2 & 3 -

- This Sheet contains information about the Sales of Belts & Goggles respectively.
- This Sheet lists sales data of belts on a weekly basis.
- Total 9 weeks of data is collected
- Quantity : The total number of items sold on a particular day.

Insights -

The store is operational from 9am to 9pm. The owner keeps track of every individual item sold and this thing basically helps them to calculate their weekly or monthly sales and profits.

Information about the Inventory Data of Business - [Inventory Data](#)

Information Collected : This excel file contains 2 sheets.

→ Sheet 1 -

- This sheet contains inventory data of belts.
- Brand - The brand of belt. 7 brands are available in the shop.

- Quantity bought - The entire quantity of belts from a specific brand purchased by the owner.
- Date - The date on which the items are bought.
- Quantity left - (Quantities left after 2 weeks of purchasing them) + (old stock of same brand)

→ Sheet 2 -

- Almost everything is similar to the 1st sheet except that the data in this sheet is about goggles.

Insights -

Items are generally purchased by the owner every two weeks and items are often sold out by the end of the second week.

4 Descriptive Statistics

Purchase Data (Belts) :

Total number of purchases - 1075 units

Most Purchased Item - Leather with a total purchase of 210 units.

Average Cost per Unit - ₹ 95.18(approx.)

Total Purchase Cost - ₹ 1,02,325

Mean Quantity Purchased per Item - 154 units

Purchase Data (Goggles) :

Total number of purchases - 563 units

Most Purchased Item - Spike with a total purchase of 115 units.

Average Cost per Unit - ₹ 30.90(approx.)

Total Purchase Cost - ₹ 17,400

Mean Quantity Purchased per Item - 80 units

Sales Data (Belts) :

Total Number of Sales Records: 928 sales.

Top Selling Item: Leather with an average sale of 4 units/day.

Average Quantity Sold per Day: 17 units/day.

Mean Revenue per Day: ₹511.

** All the values are rounded off**

Mode of Quantity Sold per Item: 133 units for several items like Fashion and Lewis.

Sales Data (Goggles) :

Total Number of Sales Records: 463 sales.

Top Selling Item: Spike with an average sale of 2 units/day.

Average Quantity Sold per Day: 9 units/day.

Mean Revenue per Day: ₹198.

** All the values are rounded off**

Inventory Data (Belts) :

Item with Highest Remaining Stock: Soft Gear (71 units).

Mean Remaining Stock per Item: 24.14 units.

Inventory Data (Goggles) :

Item with Highest Remaining Stock: Rayvin(40 units).

Mean Remaining Stock per Item: 14.14 units.

5 Detailed Explanation of Analysis Process

The analysis process for the project involves a combination of quantitative and qualitative methods, each chosen for their ability to address specific aspects of the business's challenges. Extensively utilized spreadsheets and their functions for various calculations essential to the analysis process. Spreadsheets provided a familiar and user-friendly interface for conducting complex calculations and aggregating data. Functions such as SUM, AVERAGE, PIVOT TABLE etc were employed to streamline the process of summarizing and analyzing large datasets.

ABC Analysis:

In this project, ABC Analysis was employed to categorize the inventory based on the value and impact of different items on overall profit generation. I began by extracting data from the shop's purchase and sales records. Items were classified into three categories:

Category A included high-value items that contribute to roughly 70-80% of the total value but represent a small portion (e.g., 10-20%) of the total inventory. For example, Leather & Lewis belts are among the most expensive items in the inventory, but they are also highly popular with customers. Due to their strong demand, they sell quickly and generate immediate profit.

Similarly, brands like Fashion and Rexin, which are some of the most affordable options, also sell out swiftly and do not remain in inventory for long. Category A items should be prioritized for regular stock checks and restocking as they are crucial for revenue.

Category B consisted of moderately priced items like Woodland and Ajanta, which still demanded attention but did not have as high an impact as Category A items. These items account for about 15-20% of the value and 30-40% of the items. Category B items need balanced attention and can be reviewed periodically to prevent overstocking or understocking.

Category C consists of items that contribute to the remaining 5-10% of the value but make up a significant portion (e.g., 40-50%) of the inventory. For example, Soft Gear belts and 2080 and Rayvin Goggles, these items were ordered in smaller quantities to avoid overstocking but still somehow these items managed to overstock and had filled up the inventory. Category C items often have the highest potential for dead stock. We will consider strategies such as promotions, discounts, or bundle sales to clear this inventory.

Trend Analysis & Demand Forecasting:

This method is particularly suitable for financial data, which is inherently time-dependent. By examining trends, patterns, and variations over time, we can gain insights into the business's financial health and performance. This method stands out because it allows for understanding of trends based on historical data. I used sales data to identify patterns in product demand over time.

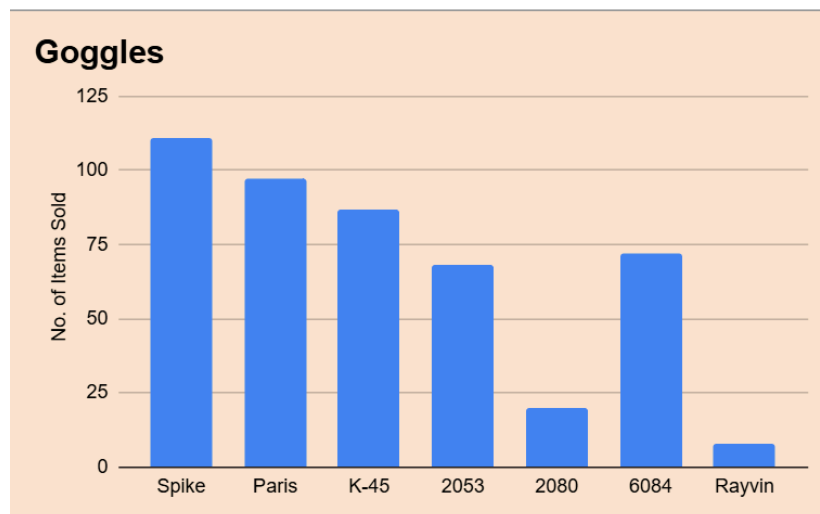
I noticed a stable demand for Leather Belts and Spike & Paris Goggles, which accounted for over 40% of sales. On the other hand, products like 2053 goggles showed more variability in sales, with some items like Lewis belts seeing occasional dips and highs. Trend analysis is also useful in predicting future demand. For example, if a product shows consistent growth, it indicates that more stock would be needed to meet future demand. This analysis helped inform more strategic purchasing decisions, allowing for better planning of stock levels and avoiding unnecessary costs related to overstocking of items like Soft Gear belts.

Conversations:

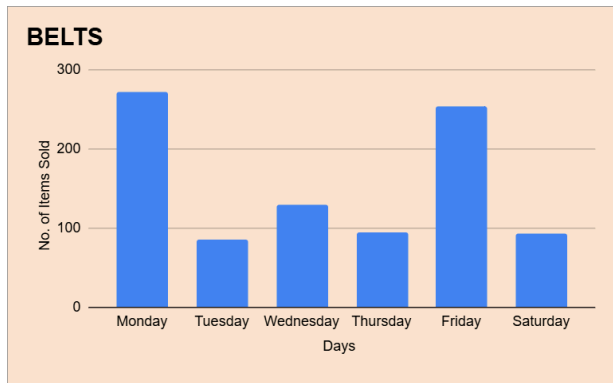
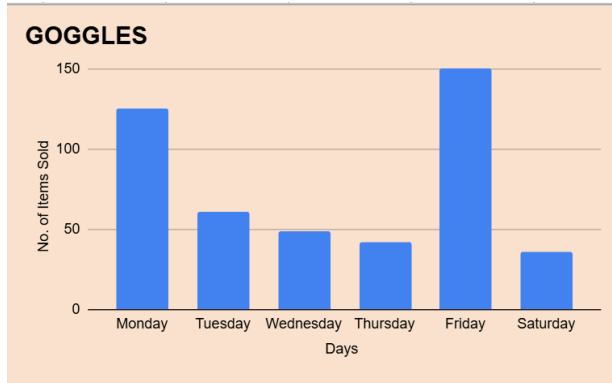
Engaging with the business owner provides qualitative insights that are not captured by quantitative data alone. Understanding the owner's perspective is also very helpful for me.

By combining these methods, I was able to develop a comprehensive strategy to improve inventory control, increase sales, and clear dead stock from the shop. The chosen methods are more appropriate than others because they offer a comprehensive view of the business's operations, combining hard data with the owner's subjective experiences. This holistic approach ensures that recommendations are not only data-driven but also grounded in the reality of the business environment. The combination of trend analysis, statistical computation, and conversations offers a balanced and thorough understanding of the business's challenges and opportunities.

6 Results and Findings



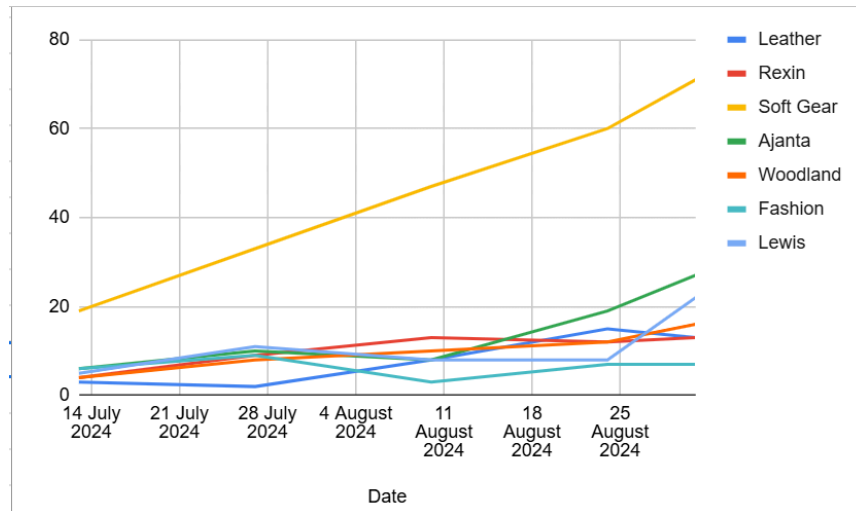
- The Bar Graph shows that Spike goggles are the top seller around the 9 weeks.
- Paris, k-45, 2053, 6084 are constant performers seeing dip in performance in some of the weeks.
- 2080 and Rayvin are the least selling goggles and this performance leads to making them a dead stock.



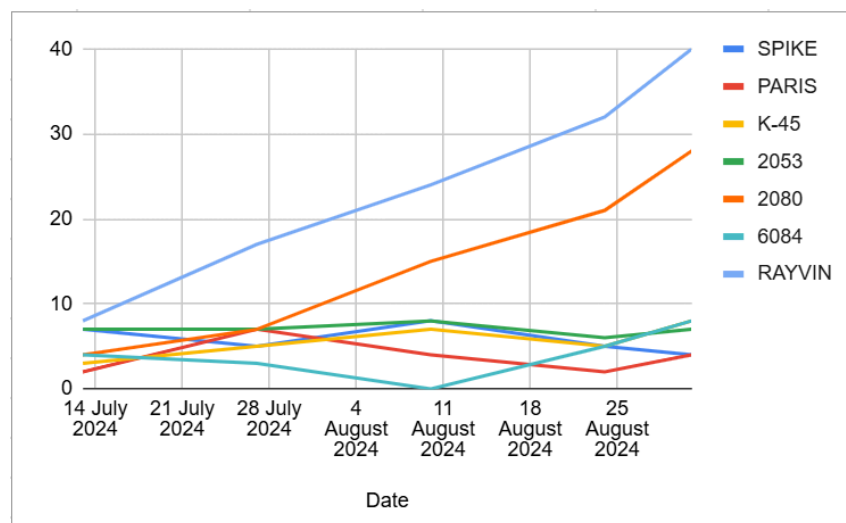
- These bar graphs shows that Mondays and Fridays are the busiest days in the market. Almost 2 to 3 times the sales are increased these days as the Monday & Friday market is held in the area where the shop is located.
- Rest of the days show constant sales which is also a good sign for a safer side of business.



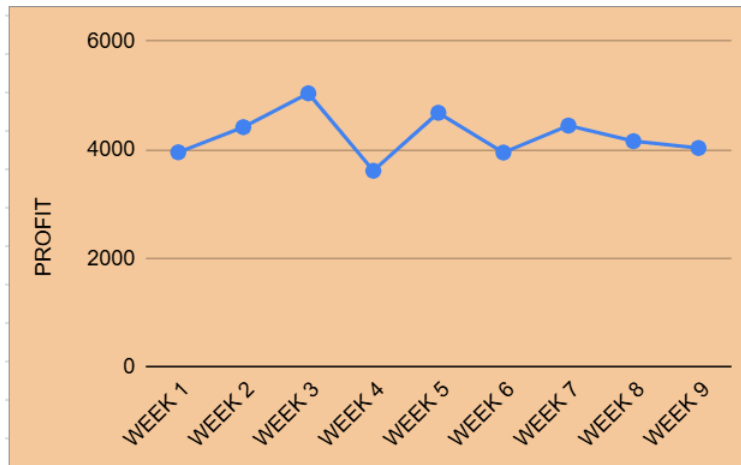
- This bar graph shows that Leather belts are the most popular belts among the customers.
- Soft Gear belts are performing very poorly in the market.
- Rest of the 5 belts Rexin , Ajanta , Woodland , Fashion , Lewis are almost having equal sales and are consistent performers in the market.



- This line chart is created through Inventory data of Belts.
- This chart shows that the line representing the Soft Gear belt is clearly increasing its number in inventory every two weeks.
- All the other lines are somewhat constant and in limit.



- This line chart is created through Inventory data of Goggles.
- This chart shows that the line representing the Spike belt is clearly increasing its number in inventory every two weeks.
- Similarly , 2080 is also becoming a dead stock gradually.
- All the other lines are constant.



- This graph shows week wise profit of the shop 'Sudhir kumar Belts & Goggles'.
- The profit remains relatively stable throughout the weeks, with no significant increases or decreases observed.
- A slight decline in profit may be observed during week 4, which could be attributed to the end of the month. A similar pattern is noticeable in week 9, which also coincides with the end of the month.